A well-order satisfies the transfinite induction schema

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted span contains no external reference (no olref inside lines 1265-1284); the existence of a $<$-minimal element of X is invoked from the well-ordering property, packaged in the same unit by the preceding proposition wo:strictorder (source lines 1250-1263); the author footnote after the proof (formulae may carry parameters) is commentary outside the counted proof.

Editorial note. The label propwoinduction names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Well-ordering minimal-element property. Same unit, the definition of a well-order together with the preceding proposition wo:strictorder (source lines 1250-1263). The proof opens the contrapositive by collecting the counterexamples into a non-empty set X and invoking that X has a $<$-minimal element; this is exactly the well-ordering property packaged by wo:strictorder (every non-empty subset has a least member).

Proposition 0.1. *If $<$ well-orders A, then for any formula $\phi(x)$:*

$$\textit{if } (\forall a \in A)((\forall b < a)\phi(b) \to \phi(a)), \textit{ then } (\forall a \in A)\phi(a).$$

Proof. We will prove the contrapositive. Suppose $\neg(\forall a \in A)\phi(a)$, i.e., that $X = \{ x \in A \mid \neg\phi(x) \} \neq \emptyset$. Then X has an $<$-minimal element, a. So $(\forall b < a)\phi(b)$ but $\neg\phi(a)$. □

This last